(T3)

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$$p(x) = \begin{cases} e^{-x/\theta}/\theta, & x \geq 0 \\ 0, & x < 0 \end{cases} \qquad \theta > 0 \quad ; \quad n = 3$$

$$\theta: \tilde{\theta}_1 = \bar{x} = \frac{1}{n}\sum_i x_i$$

$$\tilde{\theta}_3 = x_{(2)}$$

$$\boxed{M\xi = \theta} \quad \boxed{D\xi = \theta^2}$$

a) Несмещенность

1) $M\tilde{\theta}_1 = M\left(\frac{1}{n}\sum x_i\right) = \frac{1}{n} n M\xi = M\xi = \theta \Rightarrow$ несмещенная

$$M\xi = \int_{-\infty}^{+\infty} x\,dF(x) = \int_0^{+\infty} \frac{x}{\theta}\cdot e^{-x/\theta}\,dx = -\frac{\theta}{\theta}\int_0^{+\infty} x\,de^{-\frac{x}{\theta}} = -xe^{-\frac{x}{\theta}}\Big|_0^{+\infty} +$$

$$+ \int e^{-x/\theta}\,dx = -\theta e^{-\frac{x}{\theta}} = \theta \;;$$

$$M\xi^2 = \int_{-\infty}^{+\infty} x^2\,dF(x) = \frac{1}{\theta}\int_0^{+\infty} x^2 e^{-x/\theta}\,dx = -\frac{\theta}{\theta}\int_0^{+\infty} x^2\,de^{-x/\theta} = \cancel{-e^{-x/\theta}\cdot x^2\Big|_0^{+\infty}} +$$

$$0$$

$$+2\int_0^{+\infty} e^{-x/\theta}\cdot x\,dx = -2\theta\int_0^{+\infty} x\,de^{-x/\theta} = \cancel{-2\theta\, x\cdot e^{-x/\theta}\Big|_0^{+\infty}} +$$

$$0$$

$$+ 2\theta\int_0^{+\infty} e^{-x/\theta}\,dx = -2\theta^2\cdot e^{-x/\theta}\Big|_0^{+\infty} = 2\theta^2$$

$$D\xi = M\xi^2 - (M\xi)^2 = 2\theta^2 - \theta^2 = \theta^2$$

2) $\tilde{\theta}_2 = X_{(2)}$; $\xi_{(k)} \sim \sum_{i=k}^{n} C_n^i F^i(t)(1-F(t))^{n-i}$;

$$\xi_{(2)} \sim C_3^2 F^2(x)(1-F(x)) + C_3^3 F^3(x) = 3F^2(x)(1-F(x)) + F^3(x) =$$

$$= 3F^2(x) - 2F^3(x) = \psi(t); \quad \boxed{\begin{aligned} p(x) &= (e^{-x/\theta}/\theta) \\ F(t) &= (1-e^{-x/\theta}) \end{aligned}}$$

$$\psi(x) = (\psi(x))' = 6F(x)\cdot p(x) - 6F^2(x)\cdot p(x)$$

$$\psi(x) = 6\cdot(1-e^{-x/\theta})\cdot e^{-x/\theta}/\theta - 6\cdot(1-e^{-x/\theta})^2\cdot e^{-x/\theta}/\theta =$$

$$= 6(1-e^{-x/\theta})\cdot e^{-x/\theta}/\theta\cdot\left(1 - 1 + e^{-x/\theta}\right) = \frac{6(1-e^{-x/\theta})\cdot e^{-\frac{2x}{\theta}}}{\theta} =$$

$$= \boxed{6\left(\frac{e^{-\frac{2x}{\theta}}}{\theta} - \frac{e^{-\frac{3x}{\theta}}}{\theta}\right)}$$

$$M\hat{\theta}_2 = M x_{(2)} = \int_{-\infty}^{+\infty} 6x\left(\frac{e^{-\frac{2x}{\theta}}}{\theta} - \frac{e^{-\frac{3x}{\theta}}}{\theta}\right)dx = 6\int_{-\infty}^{+\infty} x\frac{e^{-\frac{2x}{\theta}}}{\theta}dx -$$

$$-6\int_{-\infty}^{+\infty} x\frac{e^{-\frac{3x}{\theta}}}{\theta}dx = -\frac{6}{\theta}\cdot\frac{\theta}{2}\int_0^{+\infty} x\,de^{-\frac{2x}{\theta}} + \frac{6}{\theta}\cdot\frac{\theta}{3}\int_0^{+\infty} x\cdot de^{-\frac{3x}{\theta}} =$$

$$= \cancel{-3x\cdot e^{-\frac{2x}{\theta}}\Big|_0^{+\infty}}_{0} + 3\int_0^{+\infty} e^{-\frac{2x}{\theta}}dx + \cancel{2x\cdot e^{-\frac{3x}{\theta}}\Big|_0^{+\infty}}_{0} - 2\int_0^{+\infty} e^{-\frac{3x}{\theta}}dx$$

$$= -3\cdot\frac{\theta}{2}e^{-\frac{2x}{\theta}}\Big|_0^{+\infty} + \frac{2\theta}{3}\cdot e^{-\frac{3x}{\theta}}\Big|_0^{+\infty} = \frac{3\theta}{2} - \frac{2\theta}{3} = \boxed{\frac{5}{6}\theta} \Rightarrow$$

$\Rightarrow$ Смещенная.

$\tilde{\theta}_2' = \frac{6}{5}\tilde{\theta}_2 \Rightarrow M\tilde{\theta}_2' = \theta$ (несмещенная)

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b) 1) $\tilde{\theta}_1$: $D\tilde{\theta}_1 = D\left(\frac{1}{n}\sum_1 x_i\right) = \frac{D\xi}{n} = \frac{\theta^2}{n} = \boxed{\frac{\theta^2}{3}}$

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2) $$M\tilde{\theta}_2^2 = 6\int_0^{+\infty} x^2\cdot\frac{e^{-\frac{2x}{\theta}}}{\theta} - 6\int_0^{+\infty} x^2\cdot\frac{e^{-\frac{3x}{\theta}}}{\theta} = -\frac{\cancel{6}^{-3}\,\cancel{\theta}}{\cancel{\theta}\cdot\cancel{2}}\int_0^{+\infty} x^2 de^{-\frac{2x}{\theta}} +$$

$$+\frac{\cancel{6}^{2}\,\cancel{\theta}}{\cancel{\theta}\cdot\cancel{3}}\int_0^{+\infty} x^2 de^{-\frac{3x}{\theta}} = -3\int_0^{+\infty} x^2 de^{-\frac{2x}{\theta}} + 2\int_0^{+\infty} x^2 de^{-\frac{3x}{\theta}} =$$

$$= \cancel{-3x^2 e^{-\frac{2x}{\theta}}\Big|_0^{+\infty}}_{0} + 6\int_0^{+\infty} x e^{-\frac{2x}{\theta}}dx + \cancel{2x^2 e^{-\frac{3x}{\theta}}\Big|_0^{+\infty}}_{0} - 4\int_0^{+\infty} x\cdot e^{-\frac{3x}{\theta}}dx =$$

$$= -\frac{6\theta}{2}\int_0^{+\infty} x\,de^{-\frac{2x}{\theta}} + \frac{4\theta}{3}\int_0^{+\infty} x\,de^{-\frac{3x}{\theta}} = \cancel{-\frac{6\theta}{2}x e^{-\frac{2x}{\theta}}\Big|_0^{+\infty}}_{0} + 3\theta\int e^{-\frac{2x}{\theta}}dx$$

$$\cancel{+\frac{4}{3}\theta x\cdot e^{-\frac{3x}{\theta}}\Big|_0^{+\infty}}_{0} - \frac{4}{3}\theta\cdot\int_0^{+\infty} e^{-\frac{3x}{\theta}}dx = -\frac{3\theta^2}{2}\cdot e^{-\frac{2x}{\theta}}\Big|_0^{+\infty} +$$

$$+\frac{4}{9}\theta^2 e^{-\frac{3x}{\theta}}\Big|_0^{+\infty} = \frac{3\theta^3}{2} - \frac{4}{9}\theta^2 = \boxed{\frac{19}{18}\theta^2} = M\tilde{\theta}_2^2$$

$$D\tilde{\theta}_2 = M\tilde{\theta}_2^2 - (M\tilde{\theta}_2)^2 = \frac{15}{18}\theta^2 - \frac{25}{36}\theta^2 = \frac{13}{36}\theta^2$$

$$D\tilde{\theta}_2' = M\tilde{\theta}_2'^2 - (M\tilde{\theta}_2')^2 = \left(\frac{6}{5}\right)^2 \cdot D\tilde{\theta}_2 = \frac{\cancel{36}^{1}}{25}\cdot\frac{13}{\cancel{36}_{1}}\theta^2 = \boxed{\frac{13}{25}\theta^2}$$

$D\tilde{\theta}_1 = \frac{\theta^2}{3}$ $\qquad$ $D\tilde{\theta}_1 < D\tilde{\theta}_2' \Rightarrow \tilde{\theta}_1$ <u>эффективнее</u> $\tilde{\theta}_2$

$D\tilde{\theta}_2' = \frac{13}{25}\theta^2$ $\qquad$ $\forall\theta \in \Theta \subset (0;+\infty)$

c) Нер-во Крамера-Рао (условия):

(1) Регулярность модели

(2) Регулярность оценки.

(1) $\rightarrow dF(x,\theta),\ \theta\in\Theta\subset R^1$; $F(x,\theta) = (1-e^{-x/\theta})\{(0)+\infty)\}$ –

– непрер. дифф. на $\theta \in \Theta \subset R^1$

$\rightarrow$ б) $\frac{\partial}{\partial\theta}\int_{-\infty}^{+\infty} dF(x,\theta) = \int_{-\infty}^{+\infty}\frac{\partial F(x,\theta)}{\partial\theta} \Rightarrow \int\frac{\partial p(x,\theta)}{\partial\theta}dx = 0$

$$\int_{\theta}^{+\infty}\frac{\partial\left(e^{-x/\theta}/\theta\right)}{\partial\theta}dx = \int_0^{+\infty}\frac{x\cdot e^{-x/\theta}}{\theta^3} - \int_0^{+\infty}\frac{e^{-x/\theta}}{\theta^2}dx = \frac{1}{\theta} - \frac{1}{\theta} = \underline{0}$$

$\rightarrow$ <u>Информация Фишера:</u> $I(\theta) = M\left[\left(\frac{\partial \ln p(x,\theta)}{\partial\theta}\right)^2\right] =$

$$= \int_0^{\infty}\left(\frac{x}{\theta^2}-\frac{1}{\theta}\right)^2\cdot\frac{e^{-\frac{x}{\theta}}}{\theta}dx = \int_0^{\infty}\frac{x^2}{\theta^4}\cdot\frac{e^{-x/\theta}}{\theta}dx - 2\int_0^{\infty}\frac{x}{\theta^3}\cdot\frac{e^{-\frac{x}{\theta}}}{\theta}dx +$$

$$+\int_0^{\infty}\frac{1}{\theta^2}\cdot\frac{e^{-\frac{x}{\theta}}}{\theta}dx = \frac{2}{\theta^2} - \frac{2}{\theta^2} + \frac{1}{\theta^2} = \frac{1}{\theta^2} = I(\theta) > 0 \quad \forall\theta\in\Theta\subset R$$

непрер.

$\Rightarrow$ <u>Модель регулярна</u>

(2) Оценка регулярна

а) $\tilde{\theta}_1 \rightarrow$ несмещенная, модель регулярная, $D\tilde{\theta}_1 = \frac{\theta^2}{3}$ – огр на $\forall$ компакте $\theta\in\Theta$

$\Rightarrow$ по достат. усл $\Rightarrow$ регулярная оценка.

б) $\tilde{\theta}_2' \rightarrow$ несмещ., модель регулярная, $D\tilde{\theta}_2' = \frac{13}{25}\theta^2$ – огр. на $\forall$ комп. $\theta\in\Theta$

$\Rightarrow$ оценка $\tilde{\theta}_2'$ – регулярная

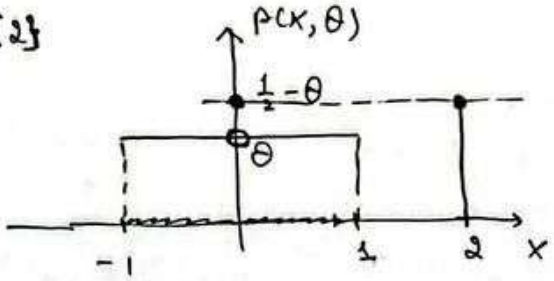
Нер-ва Кромера-Рао работает для обеих оценок.

а) $D\tilde{\theta}_1 = \frac{\theta^2}{3} = \frac{1}{n\frac{1}{\theta^2}} = \frac{\theta^2}{3}$ => по достаточн. усл. оценка эффективна

б) $D\tilde{\theta}_2' = \frac{13}{25}\theta^2 > \frac{\theta^2}{3}$ – ничего не можем сказать.

Но из ① о единственности эфф. оценки => $\tilde{\theta}_2'$ – не эффективная оценка.

$\boxed{T4}$ $g(x) = a\{(-1;1)\setminus\{0\}\} + b\{0\} + b\{2\}$



$$\int_{+\infty}^{-\infty} dF(x) = \int_{-1}^{+1} a\,dx + 2b = 2a + 2b = 1$$

$a + b = \frac{1}{2}$; $a \equiv \theta$ =>

=> $b = \frac{1}{2} - \theta$ => $\theta \in (0; \frac{1}{2})$

$\xi \sim g(x) = \theta\{(-1;1)\setminus\{0\}\} + (\frac{1}{2} - \theta)\{0\} + (\frac{1}{2} - \theta)\{2\}$

а) $\alpha_1 = M\xi = \int_{-\infty}^{+\infty} x\,dF(x,\theta) = \int_{-1}^{1} \theta x\,dx + (\frac{1}{2} - \theta)\cdot 2 = \boxed{1 - 2\theta}$

$\alpha_2 = M\xi^2 = \int_{-\infty}^{+\infty} x^2 dF(x,\theta) = \int_{-1}^{1} \theta x^2 dx + (\frac{1}{2} - \theta)\cdot 4 = \frac{2}{3}\theta + 2 - 4\theta = \boxed{2 - \frac{10}{3}\theta}$

$D\xi = M\xi^2 - (M\xi)^2 = 2 - \frac{10}{3}\theta - 1 + 4\theta - 4\theta^2 = \boxed{1 + \frac{2}{3}\theta - 4\theta^2}$

$\alpha_1(\theta) = \tilde{\alpha}_1$; $1 - 2\theta = \overline{x}$ => $\tilde{\theta} = \frac{(1 - \overline{x})}{2}$ **ОММ**

Исслед. на несмещ., состоят. и эффективность

1) Несмещ. $M\tilde{\theta} = M[\frac{1}{2}(1 - \overline{x})] = \frac{1}{2} - \frac{M\overline{x}}{2} = \frac{1}{2} - \frac{1 - 2\theta}{2} = \theta$ ✓ несмещ.

2) Сост. $D\overline{\theta} = \frac{1}{4} D\overline{x} = \frac{1}{4}\cdot\frac{D\xi}{n} = \frac{1 + \frac{2}{3}\theta - 4\theta^2}{4n} \xrightarrow[n\to\infty]{} 0$ => состоят. по достат. усл.

ОМП:

$L(\theta) = \prod_{i=1}^{n} g(x_i, \theta)$ – функ-ия правдоподобия : m – кол-во $\{0\}$ и $\{2\}$

$n - m$ – кол-во у $(-1;0) \cup (0;1)$

$L(\theta) = \theta^{n-m} \cdot (\frac{1}{2} - \theta)^m$; $\ln L(\theta) = (n-m)\ln\theta + m\ln(\frac{1}{2} - \theta)$

$(\ln L)' = \frac{n-m}{\theta} - \frac{m}{\frac{1}{2} - \theta} \Rightarrow (n-m)(\frac{1}{2} - \theta) - m\theta = 0$

$(\ln L)'' = -\frac{(n-m)}{\theta^2} - \frac{m}{(\frac{1}{2} - \theta)^2} < 0$

$\frac{1}{2}n - n\theta - \frac{1}{2}m = 0$

$\tilde{\theta}_2 = \frac{n-m}{2n} = \frac{1}{2} - \frac{\nu}{2}$; $\nu = \frac{m}{n}$ ~ **Оцен. ММП.**

=> точка максимума.

Исследование на несмещ., состоят.

1) $M\tilde{\theta}_2 = M[\frac{1}{2} - \frac{1}{2}\hat{p}] = \frac{1}{2} - \frac{1}{2}p = \frac{1}{2} - \frac{1}{2} + \frac{2\theta}{\theta} = \theta$ ✓ несмещ.

$\boxed{p = M\nu}$

2) $D\tilde{\theta}_2 = \frac{1}{4} D\nu = \frac{p(1-p)}{n} = \frac{(1 - 2\theta)2\theta}{4n} \xrightarrow[n\to\infty]{} 0$ ✓ состоят по достат. условию.

c) Исследование на эффективность

(1) Регулярность модели

① $p(x,\theta) = \theta\{(-1;1)\setminus\{0\}\} + (\frac{1}{2}-\theta)\{0\} + (\frac{1}{2}-\theta)\{2\}$ - непрер. дифф. по $\theta \in \Xi : (0; 1/2)$ ✓

② $\int_{-1}^{1} \frac{\partial g(x,\theta)}{\partial\theta} + \sum \beta_k(\theta) = 0$? $\int_{-1}^{1} dx + (-2) = 0$ ✓

③ $I(\theta) = M\left[\left(\frac{\partial \ln g(x)}{\partial\theta}\right)^2\right] = \int_{-1}^{1} \frac{1}{\theta^2}\theta dx + \left(-\frac{1}{\frac{1}{2}-\theta}\right)^2(1-2\theta) =$

$= \frac{2}{\theta} + \frac{4}{1-2\theta} = \frac{2}{\theta(1-2\theta)}$ => $I(\theta) > 0$ и непрер на $\theta \in (0; \frac{1}{2})$ ✓

=> Модель регулярна

(2) Регулярность оценок

$\tilde{\theta}_1$ - несмещ., модель регулярная, $D\tilde{\theta}_1 = \frac{1+\frac{2}{3}\theta+4\theta^2}{4n}$ - огр на $\forall$ изм. из $\theta \in \Xi : (0; 1/2)$

=> оценка регулярна

$\tilde{\theta}_2$ - несмещ., модель регулярная, $D\tilde{\theta}_2 = \frac{(1-2\theta)2\theta}{4n}$ - огр на $\forall$ изм из $\theta \in \Xi : (0; 1/2)$

=> оценка регулярна

=> Можно применить нер-во Крамера-Рао ! (для обеих оценок)

$\tilde{\theta}_1$: $\frac{1+\frac{2}{3}\theta-4\theta^2}{4n} \overset{?}{>} \frac{\theta(1-2\theta)}{n\cdot 2}$; $1+\frac{2}{3}\theta-4\theta^2 \overset{?}{>} 2\theta-4\theta^2$

$1+\frac{2}{3}\theta \overset{?}{>} 2\theta$

$1-\frac{4}{3}\theta \overset{?}{>} 0 \Leftarrow \theta \in (0; \frac{1}{2})$

=> Ничего сказать нельзя

$\tilde{\theta}_2$: $\frac{(1-2\theta)\cancel{2}\theta}{\cancel{4}_2 n} = \frac{\theta(1-2\theta)}{2n}$ => по достаточному условию $\tilde{\theta}_2$ - эффективна

=> из ① о единственности эфф. оценки => $\tilde{\theta}_1$ - не эффективная